TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Pressure Coefficient of a Gas by Constant Volume Air Thermometer

Trap a fixed volume of dry air, warm it, and its pressure grows in strict proportion to the temperature. The constant of that proportion — and the temperature where the line hits zero pressure — is the business of this experiment.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 19 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the pressure coefficient of air at constant volume, β, using a constant-volume air thermometer. For a fixed mass of gas held at constant volume, the pressure rises linearly with temperature (Charles's law in its pressure form):

$$p_t = p_0 (1 + \beta\, t)\,, \quad \beta = (p_t - p_0) / (p_0\, t)$$

where p_0 is the pressure at 0 °C. For an ideal gas $\beta = 1/273.15 \approx 3.66 \times 10^{-3}\ °C^{-1}$; real dry air comes very close. Extrapolating the straight line p_t versus t to zero pressure gives absolute zero, $t = -1/\beta$.

The gas sits in a glass bulb connected by a capillary to a mercury manometer. Heating makes the gas expand and push the mercury down on the bulb side; the movable reservoir is raised until the mercury meniscus returns exactly to the fiducial mark A on the capillary — that restores the volume. The pressure is then read from the manometer:

$$p = H + h$$

where H is the barometric height and h is the difference between the mercury level in the open limb and the meniscus at A.

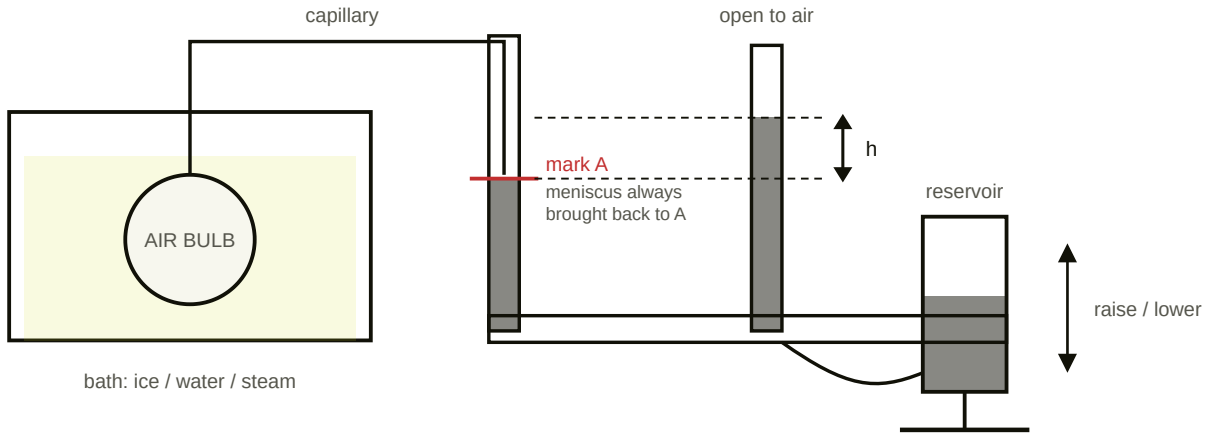


Figure 1 — The constant-volume air thermometer. The bulb stands in the bath; on heating, the reservoir is raised until the mercury meniscus returns to the fiducial mark A, keeping the gas volume constant. The pressure is then H + h, with H the barometric height.

2. Apparatus and procedure

- **Apparatus:** constant-volume air thermometer (bulb, capillary with fiducial mark, mercury manometer with movable reservoir), ice bath, water bath with heater, steam jacket or boiling bath, thermometer, barometer, cathetometer or scale for reading levels.
- Fill and adjust so that with the bulb in melting ice the mercury meniscus can be set exactly at mark A with the reservoir at a comfortable height. Read both mercury levels; their difference h_0 gives $p_0 = H + h_0$.
- Replace the ice bath with the water bath at about 50 °C (read on the thermometer). Wait for equilibrium — the meniscus stops creeping — then raise the reservoir to bring the meniscus back to A, and read the new difference h.
- Repeat with the bulb in steam (100 °C), again waiting for a steady state before setting the meniscus to A and reading h.
- Read the barometer H once for the session (it changes negligibly). Tabulate p = H + h against t, plot the graph, and compute β from the ice and steam points and from the graph's slope.

3. Worked calculation

Barometric height H = 76.0 cm Hg throughout.

BATH	T (°C)	H (CM HG)	P = H + H (CM HG)
Melting ice	0	0.0	76.0

Water bath	50	14.0	90.0
Steam	100	27.9	103.9

$$\beta = (103.9 - 76.0) / (76.0 \times 100) = 27.9 / 7600 \approx 3.67 \times 10^{-3}\ ^\circ\text{C}^{-1}$$

The 50 °C point checks the line: predicted p = 76.0 × (1 + 3.67 × 10^{-3} × 50) = 90.0 cm Hg — exactly the measured value, so the three points are collinear as the law demands. Extrapolation gives absolute zero at $-1/\beta \approx -272$ °C.

RESULT

β(air) ≈ 3.67 × 10^{-3} °C^{-1}, against the ideal-gas value 3.66 × 10^{-3} °C^{-1} (1/273.15). The small excess is typical of a real gas measured over a range where some air also sits in the cooler capillary outside the bath (the "dead space"), which slightly distorts the simple law.

4. Precautions

- The air in the bulb must be dry — moisture would add its own vapour pressure, which does not follow the simple law, especially near 100 °C. A drying tube on the capillary guards this.
- Wait for true equilibrium at each bath: the meniscus must be perfectly steady before the reservoir is adjusted to mark A; a still-warming bulb reads low.
- Set the meniscus to mark A exactly, reading at eye level to avoid parallax, and read both levels in the same posture; a 1 mm error in h is a direct error in p.
- Keep the capillary and manometer at a steady room temperature and handle the reservoir by its stand — warming the connecting tube with a hand changes the "constant" volume's surroundings.
- Bring the meniscus to A by the same approach (raising the reservoir) each time, so any sticking of the mercury in the tube affects all readings alike.

Practice (viva preparation)

1. Why must the mercury be brought back to mark A before the pressure is read?
2. With β = 3.67 × 10^{-3} °C^{-1} and p_0 = 76.0 cm Hg, predict the pressure at room temperature 27 °C.
3. What temperature does the experiment assign to absolute zero, and how is it obtained from the p–t graph?
4. The bulb is accidentally left with moist air inside. Which reading is corrupted most, the ice point or the steam point, and why?
5. Why is a large bulb and a narrow capillary a good design?

Answer key

1. The law $p_t = p_0(1 + \beta t)$ holds only at constant volume. Heating at first increases the volume (the mercury is pushed back); raising the reservoir squeezes the gas back to exactly the volume marked by A, so the pressures at different temperatures are comparable.
2. $p = 76.0 \times (1 + 3.67 \times 10^{-3} \times 27) = 76.0 \times 1.0991 \approx 83.5$ cm Hg (so $h \approx 7.5$ cm).
3. About −272 °C. The straight line through the measured points is extended to $p = 0$; its temperature intercept is $-1/\beta$, the temperature at which the gas pressure would vanish — absolute zero (ideal value −273.15 °C, reached exactly only by an ideal gas).
4. The steam point. At 0 °C the saturation vapour pressure of water is only about 0.5 cm Hg, but at 100 °C it is a full atmosphere; evaporating moisture adds a large extra pressure at the hot end that has nothing to do with the air's expansion, inflating β.
5. A large bulb means most of the gas is truly at the bath temperature and takes a big share of any change, while a narrow capillary keeps the "dead space" of gas outside the bath small and makes the meniscus position — hence the volume — precise to set and read.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib